

Business reflection questions

1. Which factors appear to influence subscription renewal the most?

The factors that commonly influence subscription renewal are:

- Customer engagement and usage frequency
- Customer satisfaction and feedback ratings
- Subscription duration or membership tenure
- Monthly spending patterns
- Number of services/features used
- Customer support experience
- Promotional offers and discounts

Customers who actively use the platform and derive value from it are generally more likely to renew their subscriptions.

2. Why is subscription renewal a classification problem?

Subscription renewal is a classification problem because the target variable belongs to one of two categories:

- Renewed (Yes)
- Not Renewed (No)

The model predicts which class a customer belongs to. Since the output is categorical rather than numerical, it is treated as a classification task.

3. Why is monthly spending a regression problem?

Monthly spending is a regression problem because the target variable is a continuous numerical value.

For example, a customer may spend ₹500, ₹1200, or ₹3500 per month. Since the model predicts an exact numerical amount rather than a category, regression algorithms are used.

4. Which algorithm performed better for renewal prediction?

The algorithm with the higher evaluation metrics (such as Accuracy, Precision, Recall, and F1-Score) performed better for renewal prediction.

Typically, if comparing common models:

- Random Forest often performs better due to its ability to capture complex patterns.
- Logistic Regression may perform well when relationships are simple and linear.

The final answer should be based on whichever model achieved the highest performance in your results.

5. How could the platform use these predictions to improve customer retention?

The platform can use prediction results to:

- Identify customers who are likely to cancel subscriptions.
- Offer personalized discounts and promotions.
- Send renewal reminders and engagement campaigns.
- Recommend relevant features or services.
- Improve customer support for at-risk users.
- Develop targeted retention strategies.

By acting before customers leave, the platform can increase retention rates and improve long-term revenue.